

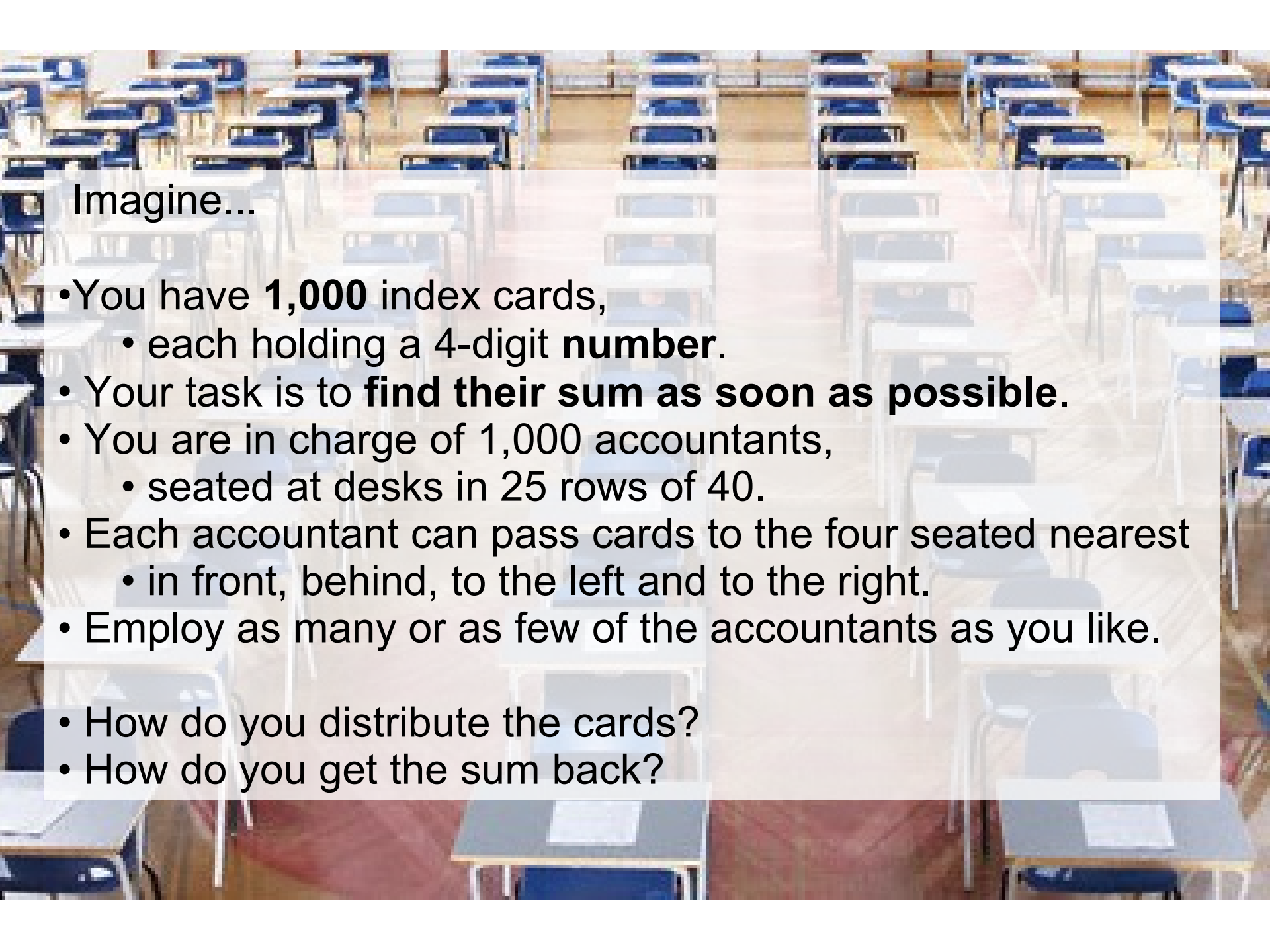
ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

Introduction to Parallel Computing

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What can parallel computing do for me?

- + Faster time-to-results
- + Bigger problems feasible
in memory
or time
- More costly
in equipment
and effort



Imagine...

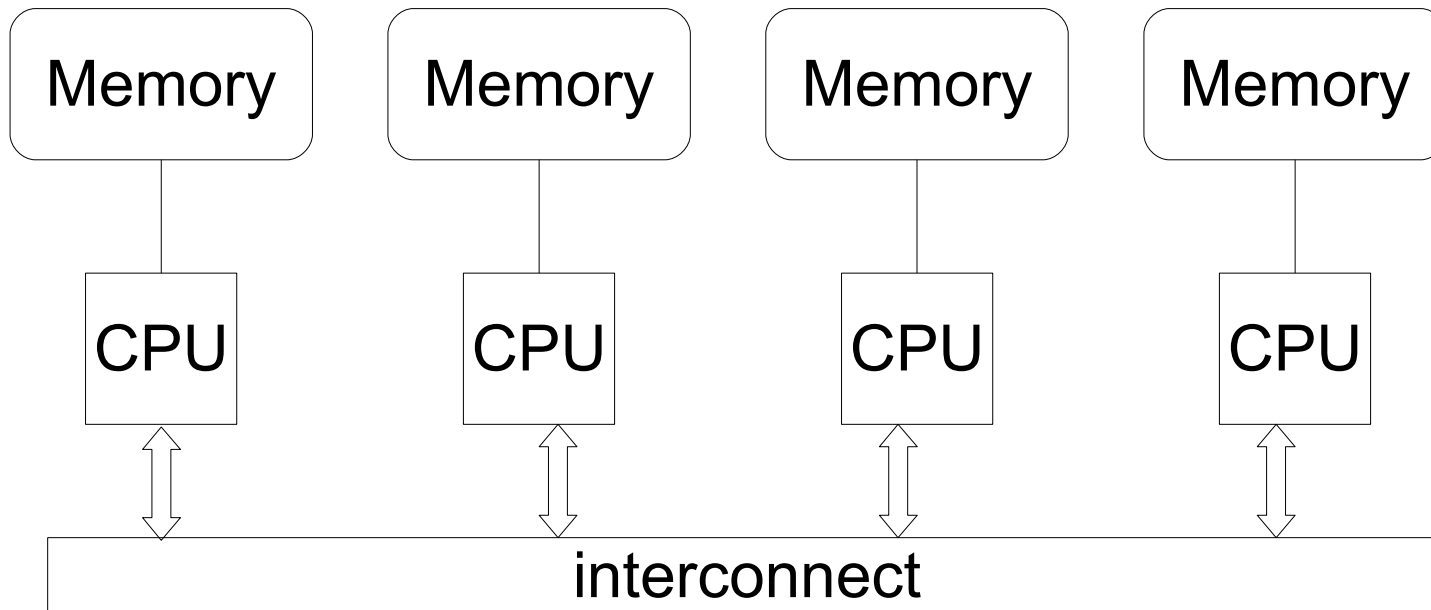
- You have **1,000** index cards,
 - each holding a 4-digit **number**.
- Your task is to **find their sum as soon as possible**.
- You are in charge of 1,000 accountants,
 - seated at desks in 25 rows of 40.
- Each accountant can pass cards to the four seated nearest
 - in front, behind, to the left and to the right.
- Employ as many or as few of the accountants as you like.
- How do you distribute the cards?
- How do you get the sum back?

Outline

- Parallel computer architectures
- Programming models
- MPI example
- OpenMP example
- Performance measurement
 - terminology

Parallel architecture:

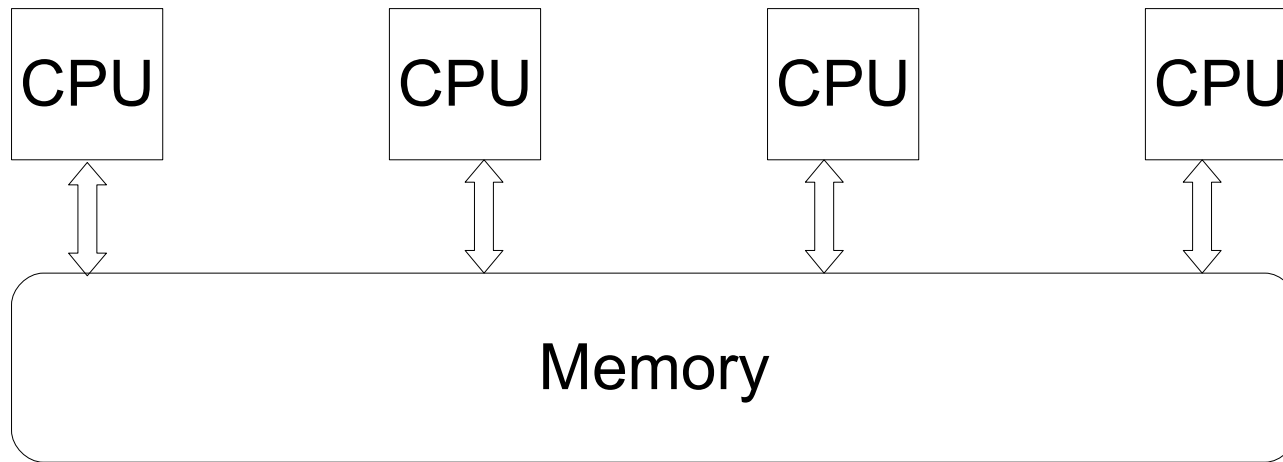
Distributed-memory



- **interconnect** can be Ethernet...
- ...or something faster and more expensive like Infiniband or Myrinet

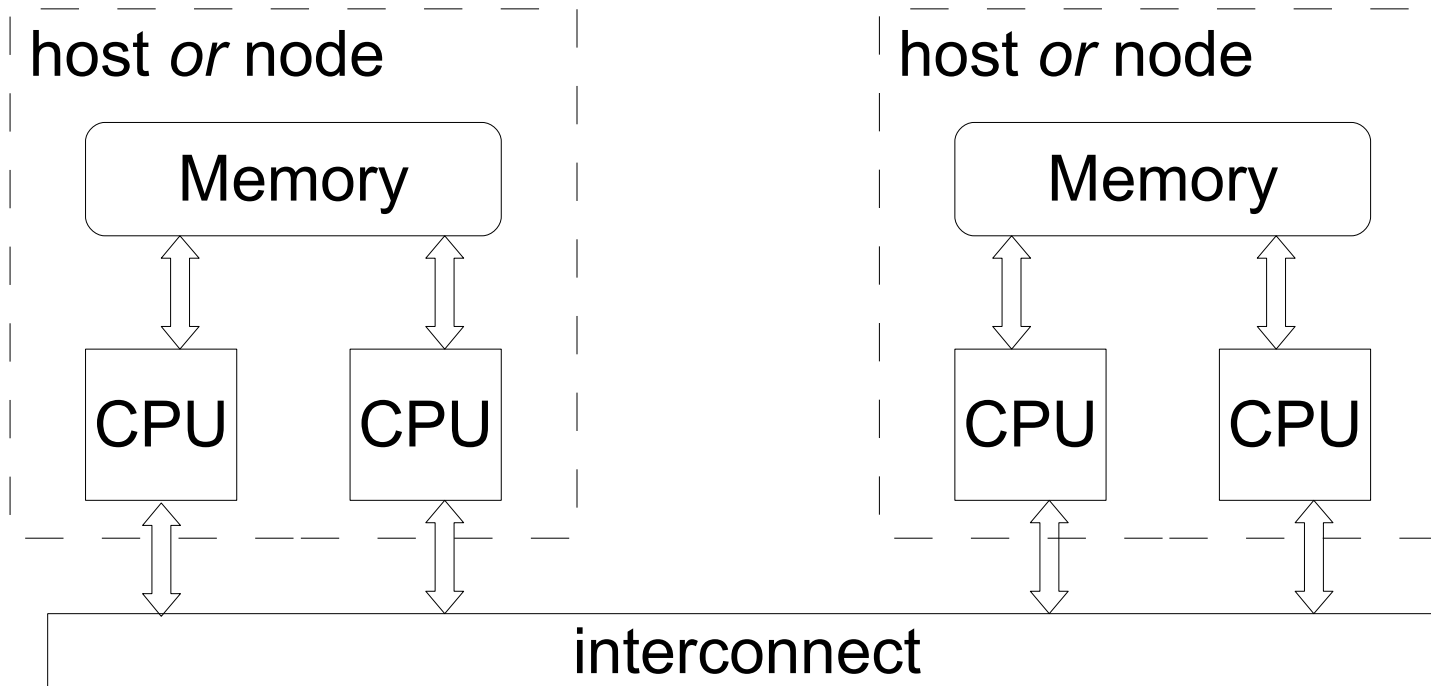
Parallel architecture:

Shared-memory



- also called “multiprocessors”
- e.g. modern dual-core or quad-core CPUs
- “cc-NUMA” a common implementation
 - *cache-consistent Non-Uniform Memory Access*

Parallel architecture: Hybrid



- Contemporary clusters usually built on this model

Programming Model 1: Independent tasks

- Near 100% efficiency
- Very simple to implement
 - handled at job-manager level
 - no changes to serial application code
- “**Embarassingly parallel**”
 - so-called because no theoretical interest
 - better term: “perfectly parallel”
- Related terms:
 - “parametric parallelism”
 - “**High-throughput computing**”

Programming Model 2:

Message-passing

- Each process has its own independent memory
 - Maps well to distributed-memory computers
- Communication between processes is explicit
 - Application has to be reprogrammed...
 - ...maybe even redesigned
- Analogous to how teams of people work
 - *“You and I don’t share memory, we pass messages” – N. Ostlund*
- Scales to arbitrary number of processors
 - ...though perhaps not efficiently!
- MPI most popular standard for message-passing programming

MPI: send-recv.c

```
#include <mpi.h>
#include <stdio.h>
int main( int argc, char *argv[] )
{ int myRank, msg=12345, source=0, destination=1, tag=99;
  MPI_Status status;

  MPI_Init(&argc,&argv);
  MPI_Comm_rank(MPI_COMM_WORLD,&myRank);

  if (myRank == source) {
    MPI_Send(&msg, 1, MPI_INT, destination, tag, MPI_COMM_WORLD);
    printf ("Process %d sent %d to process %d.\n", myRank, msg, destination);
  } else if (myRank == destination) {
    MPI_Recv(&msg, 1, MPI_INT, source, tag, MPI_COMM_WORLD, &status);
    printf ("Process %d received %d from proc %d.\n", myRank, msg, source);
  }

  MPI_Finalize();
}
```

Programming Model 3:

Shared-memory

- Each process has
 - read/write access to a **shared** memory area
 - a **private** memory area (stack)
- Maps well to shared-memory computers
 - practical limit: size of a single machine
- Communication between processes is via shared memory
 - potential for **race conditions**
- Processes sometimes called “threads”
- OpenMP and Posix Threads two popular implementations

saxpy.f95

```
program saxpy
integer, parameter :: n=10
real a,x(n),y(n)
integer i
read *,a,x(1:n),y(1:n)
```

```
do i=1,n
  y(i) = a*x(i) + y(i)
end do
```

```
print *,y(1:n)
end program
```

saxpy.f95 --- OpenMP

```
program saxpy
integer, parameter :: n=10
real a,x(n),y(n)
integer i
read *,a,x(1:n),y(1:n)

!$OMP PARALLEL PRIVATE(i) SHARED(a,x,y)
!$OMP DO
do i=1,n
    y(i) = a*x(i) + y(i)
end do
!$OMP END DO
!$OMP END PARALLEL

print *,y(1:n)
end program
```

Reduction

```
program reduce
integer, parameter :: n=10
real :: accum=0.0, x(n)
integer i
read *,x(1:n)
```

```
!$OMP PARALLEL PRIVATE(i) SHARED(x) REDUCTION(+:accum)
!$OMP DO
do i=1,n
    accum = accum + x(i)
end do
!$OMP END DO
!$OMP END PARALLEL
```

```
print *,accum
end program
```

Performance: Speedup

$$\text{Speedup} = \frac{\text{Sequential execution time}}{\text{Parallel execution time}}$$

Example:

- 6 hours to run on 1 processor (sequential)
- 1 hour to run on p processors (parallel)
- $\rightarrow 6\text{h}/1\text{h} = \text{speedup of } 6$

Performance: Speedup

$$\text{Speedup} = \frac{\text{Sequential execution time}}{\text{Parallel execution time}}$$

Consider

- the time spent in sequential work, σ
- the time spent in **parallelizable** work, φ
- parallel overhead costs, κ

...for a problem of size n running on p processors:

$$\psi(n, p) \leq \frac{\sigma(n) + \varphi(n)}{\sigma(n) + \varphi(n)/p + \kappa(n, p)}$$

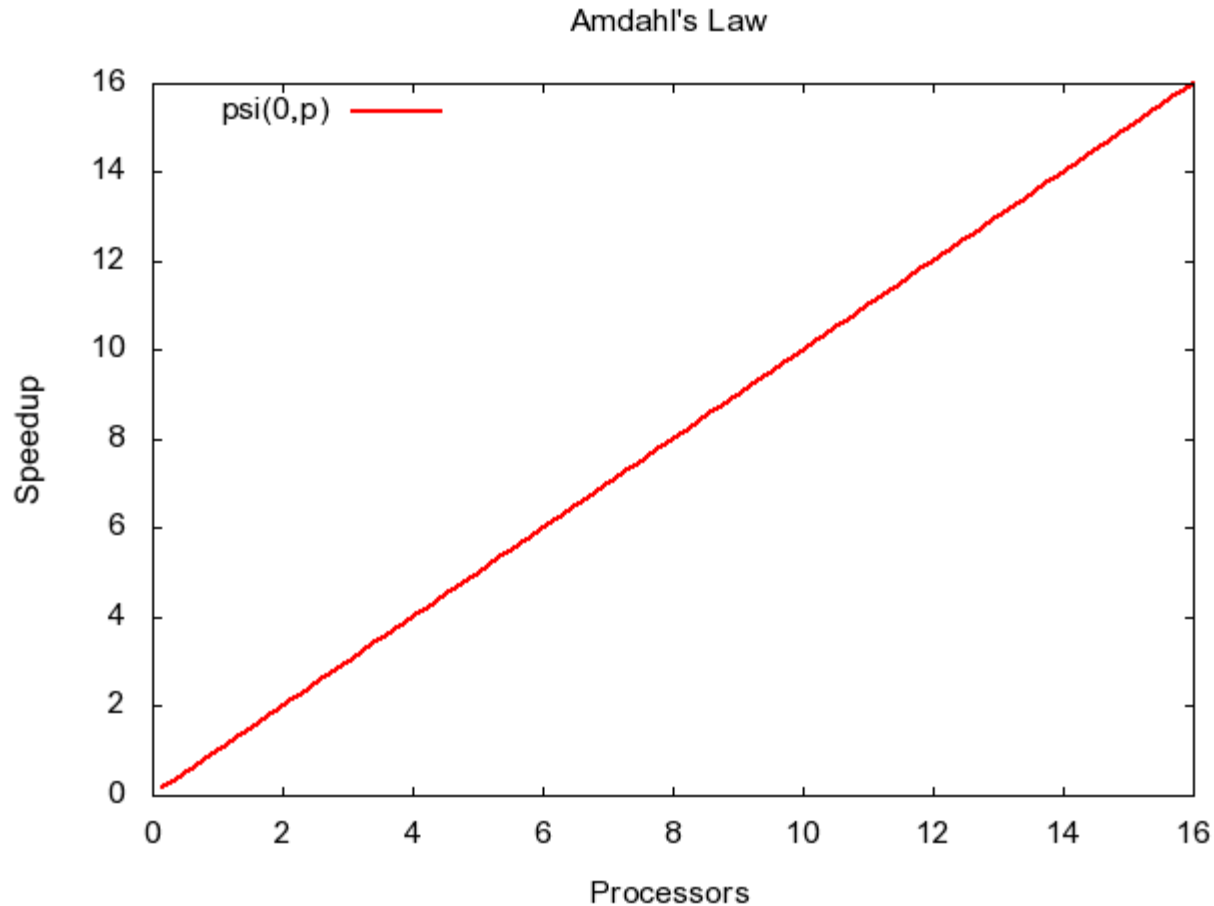
How much speedup can you get?

- Assume parallel overhead κ is negligible (Optimist!)
 - Define serial fraction $f_s = \sigma/(\sigma + \varphi)$

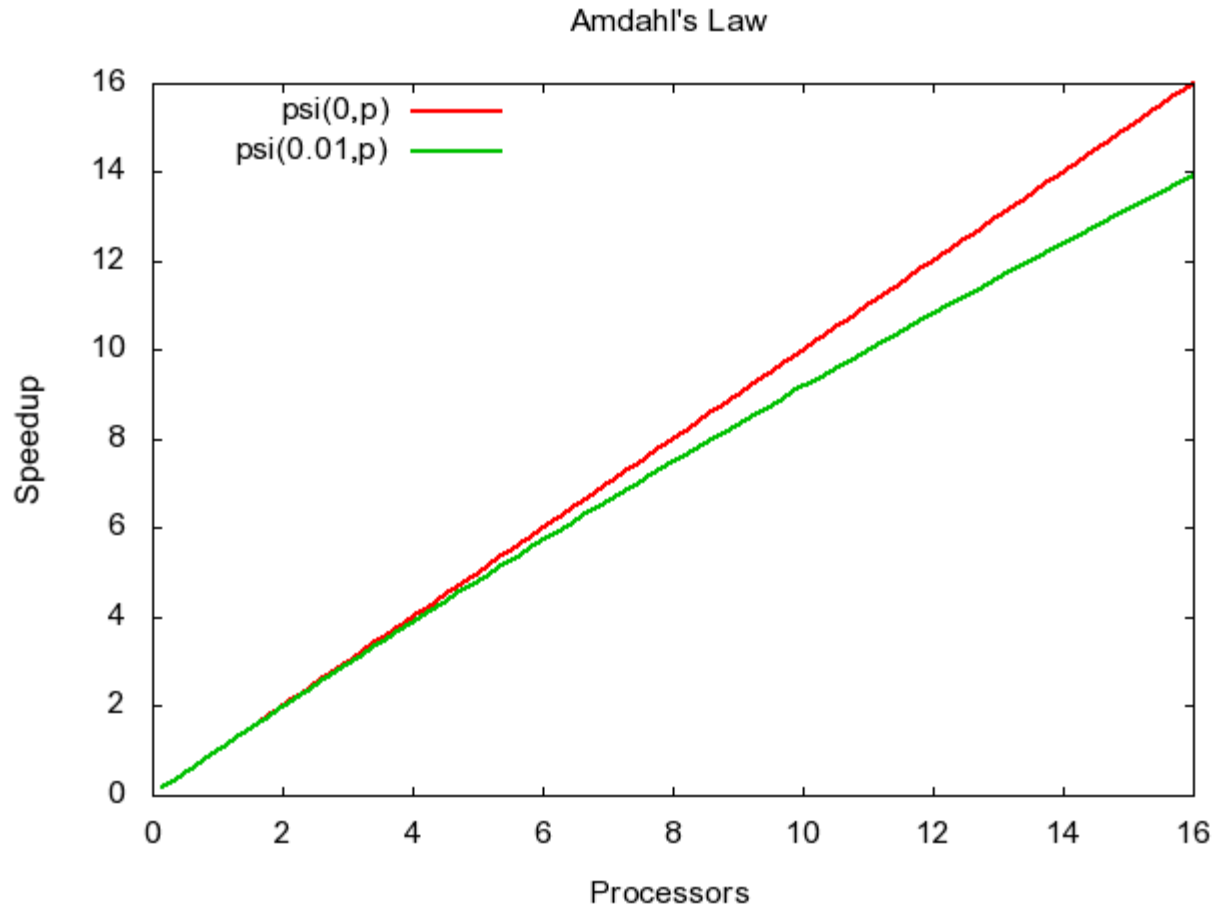
$$\psi(f_s, p) \leq \frac{1}{f_s + (1 - f_s)/p}$$

Amdahl's Law

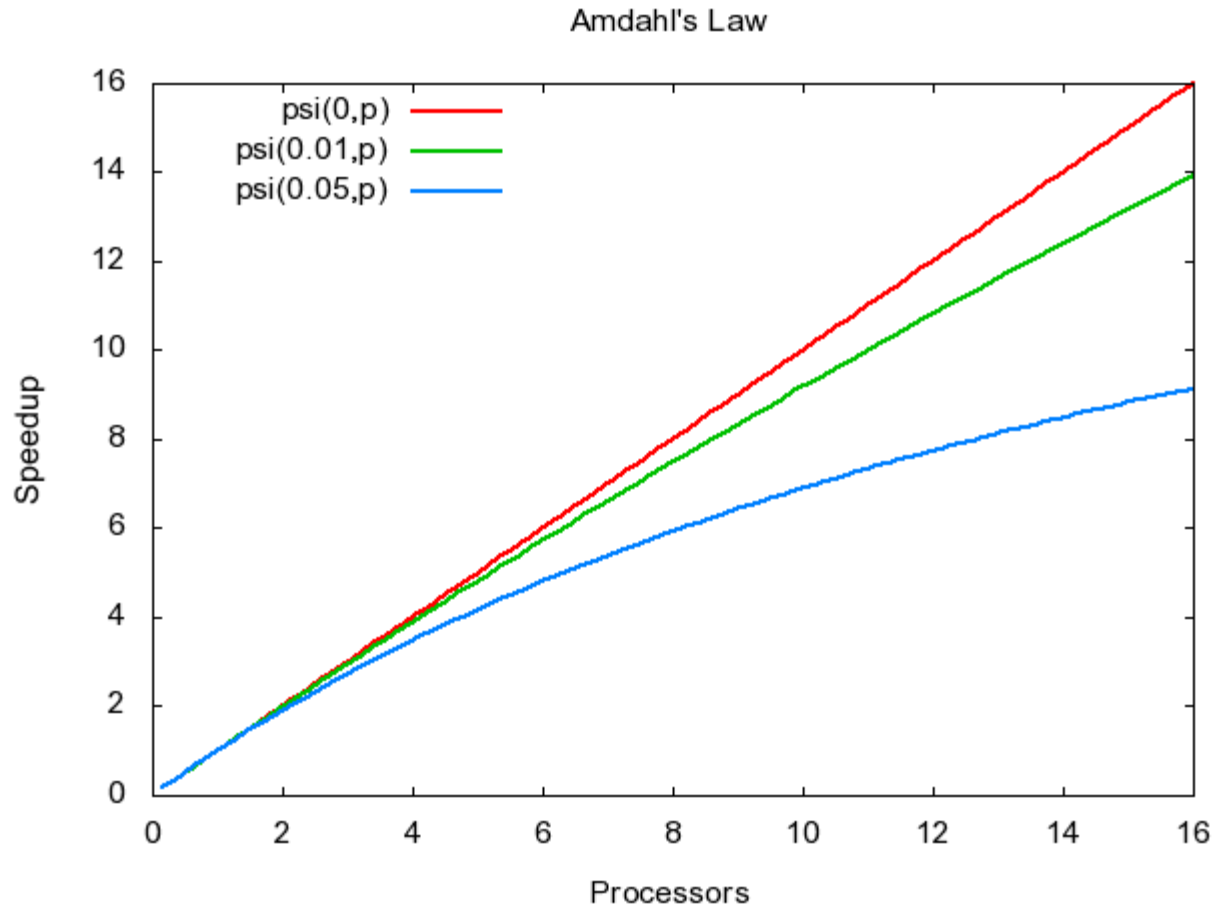
How much speedup can you get?



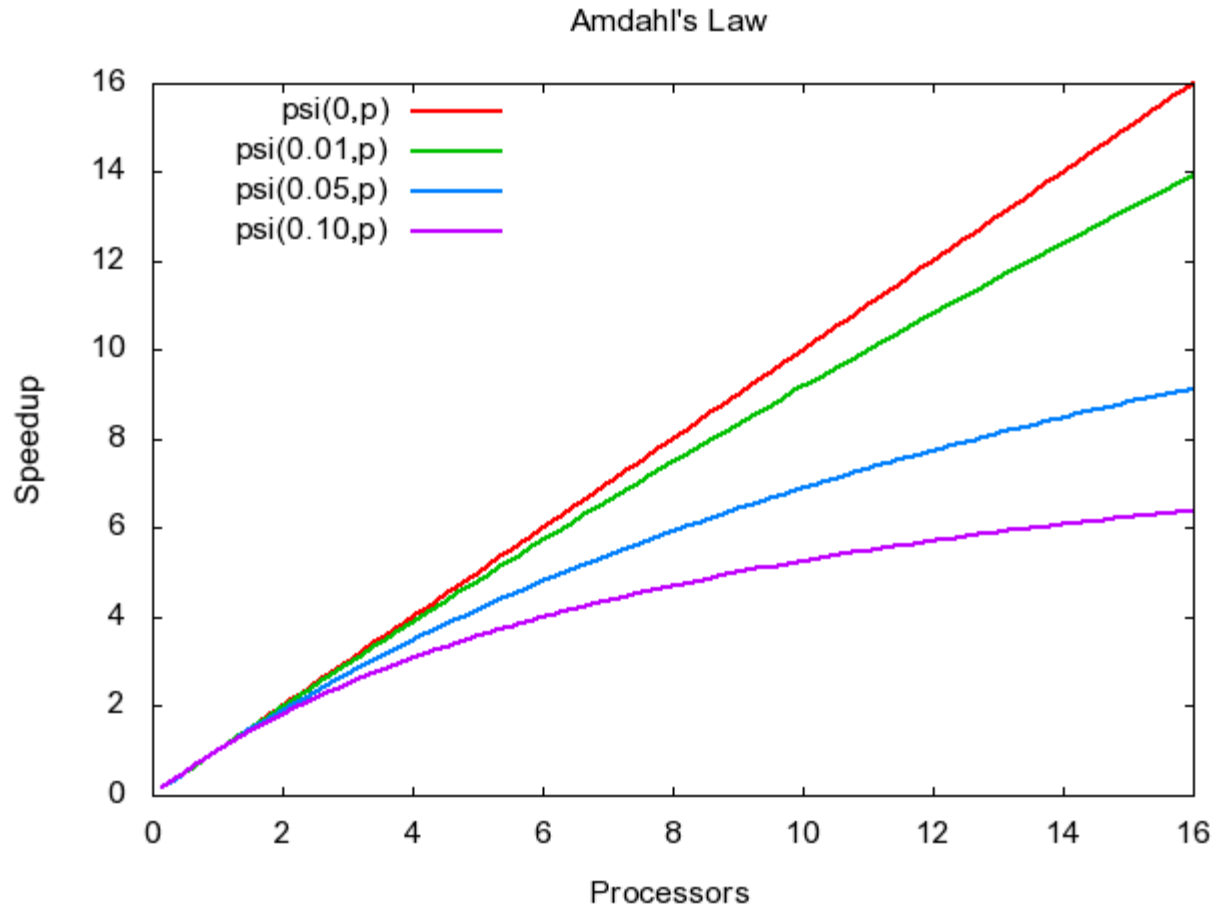
How much speedup can you get?



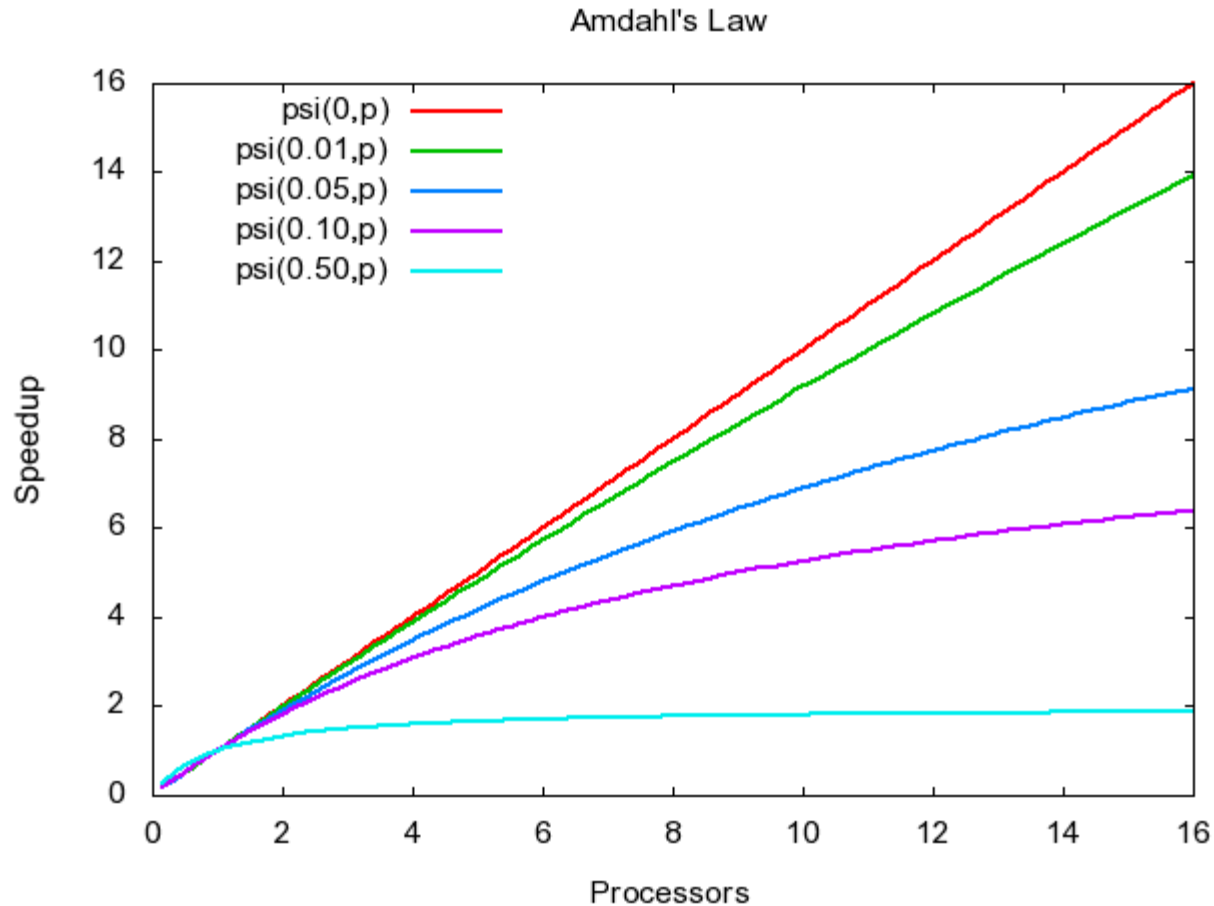
How much speedup can you get?



How much speedup can you get?



How much speedup can you get?



Efficiency

$$\text{Efficiency} = \frac{\text{Sequential time}}{\text{Processors} \times \text{Parallel time}}$$

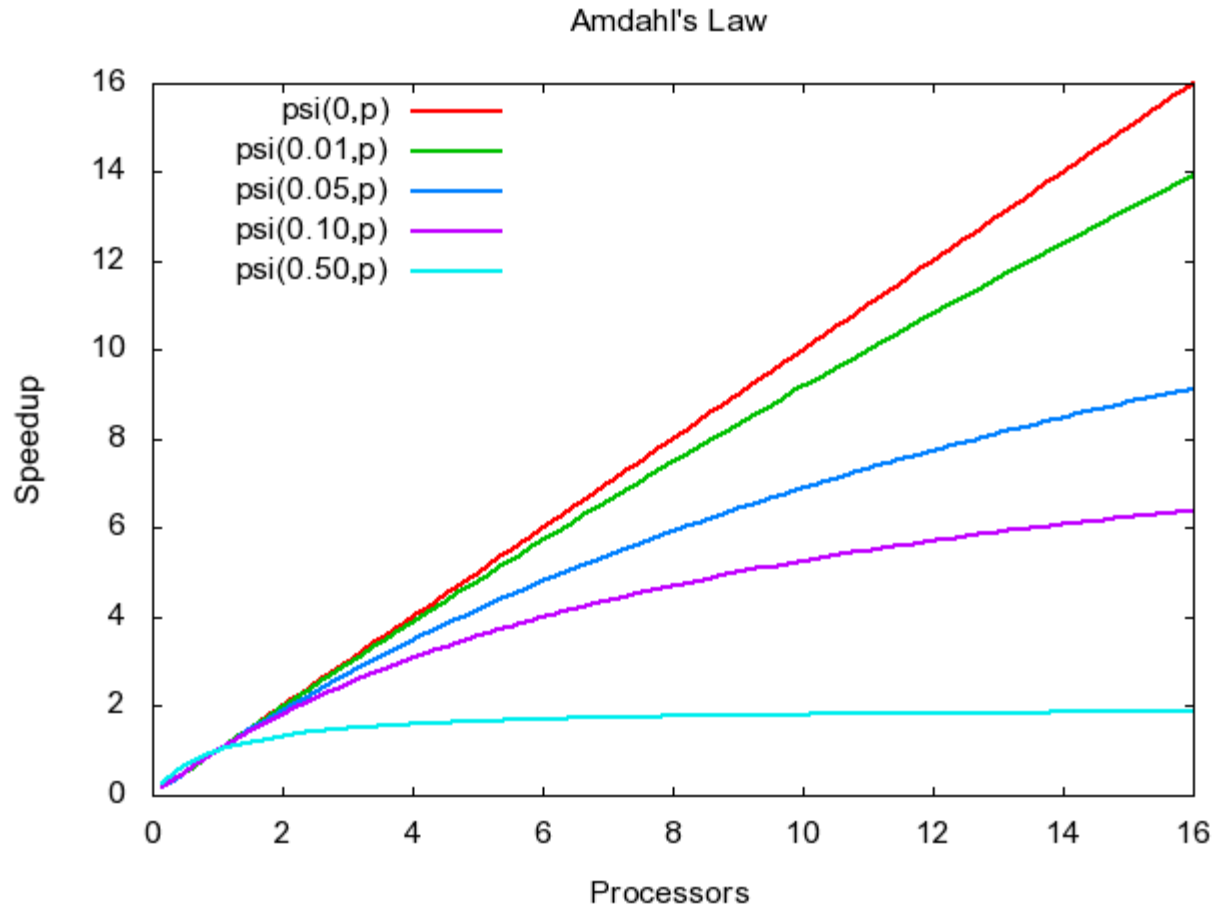
$$\text{Speedup} = \frac{\text{Sequential time}}{\text{Parallel time}}$$
$$= \frac{\text{Speedup}}{\text{Processors}}$$

- a fraction between 0 and 100%
- depends on number of processors!
- varies with size of problem

Scalability

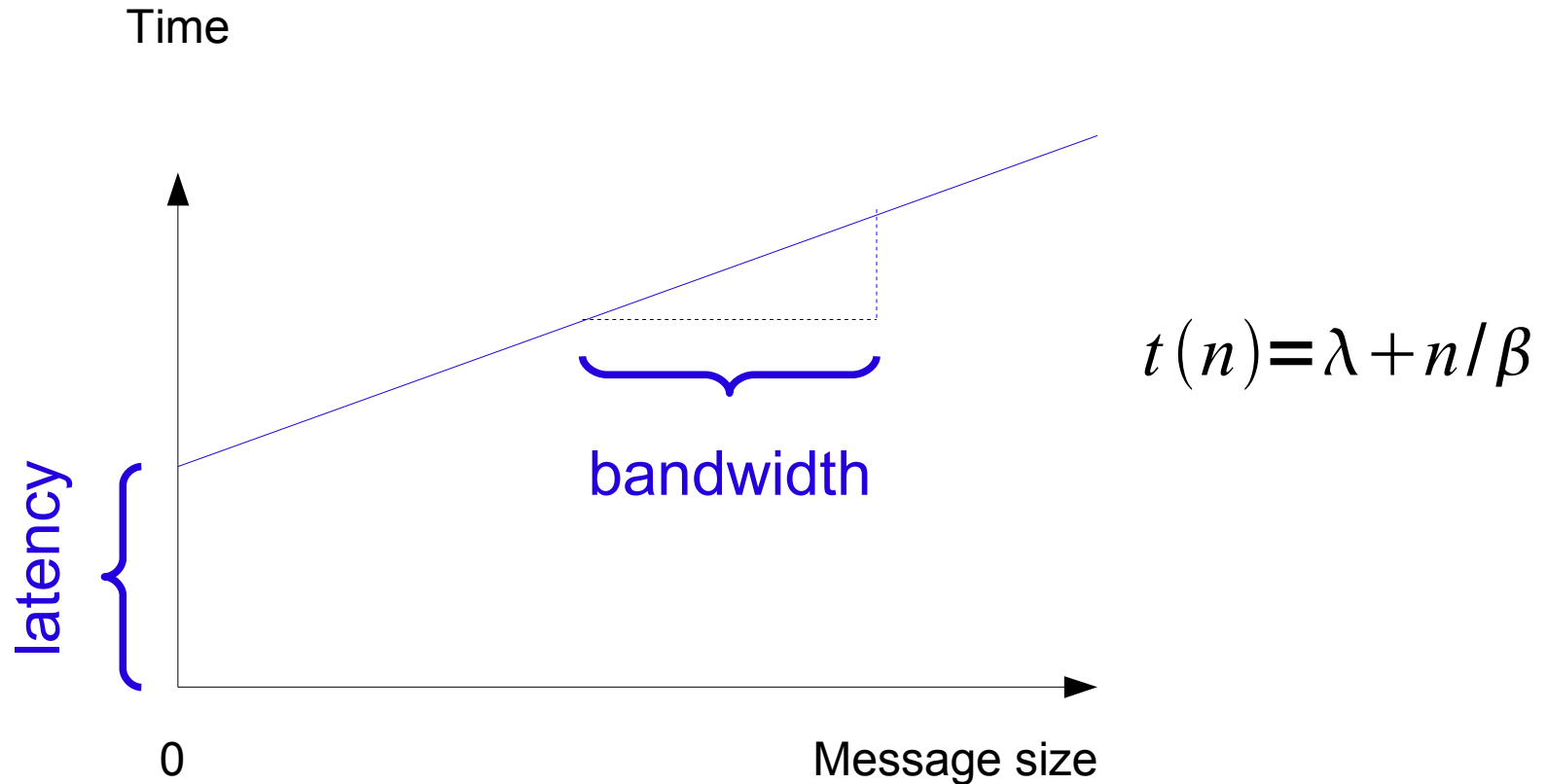
- Applies to a combination of program *and* computer
- A measure of the ability to increase performance as the number of processors increases
- Usually, as problem size increases, so does speedup!

As problem size increases, so usually does parallel fraction



Communication costs

- Passing messages takes finite time
 - So does coordinating the use of shared memory



Other resources

- Books
 - Michael J Quinn, *Parallel Programming in C with MPI and OpenMP* (ISBN 0-07-282256-2)
 - Snir et al., Gropp et al., *MPI: The Complete Reference* (2 vols.) (ISBN 978-0262692168)
 - ...many others !...
- Websites
 - <http://www.mcs.anl.gov/research/projects/mpi/>
 - <http://openmp.org/wp/>
 - ...many others !...
- ACEnet
 - <http://wiki.ace-net.ca>
 - email: support@ace-net.ca